

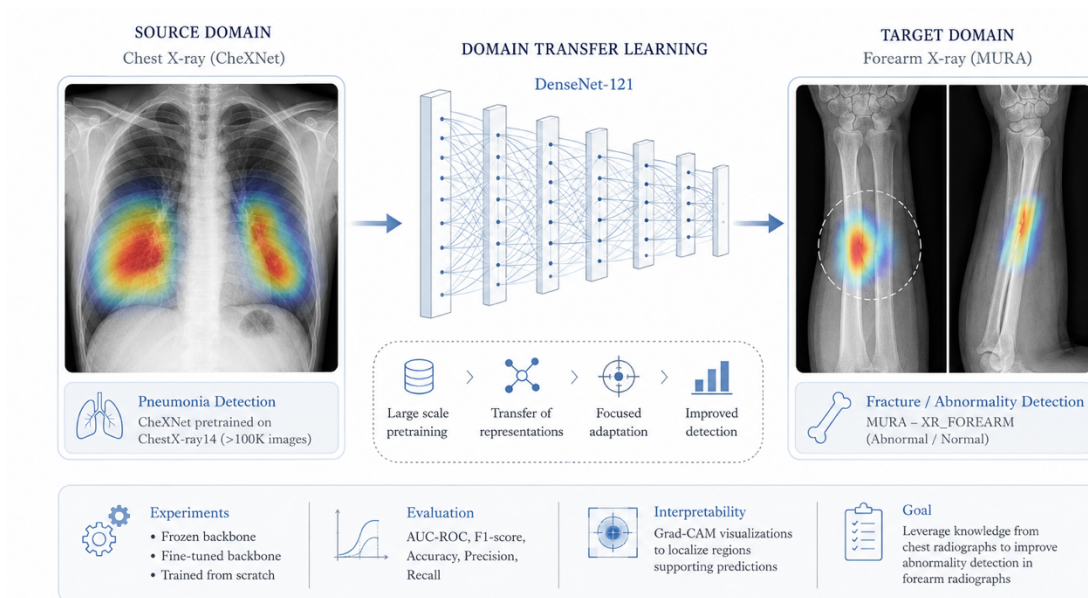
Focused Domain Transfer of CheXNet for Fracture and Abnormality Detection in Forearm Radiographs

Selected paper

CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning

Proposed extension

Transferring CheXNet's chest-X-ray representations to forearm bone radiographs (MURA) via three transfer-learning experiments, with Grad-CAM interpretability.



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1. Introduction

CheXNet is a 121-layer convolutional neural network (DenseNet-121) trained on the ChestX-ray14 dataset - over 100,000 frontal chest radiographs labelled with 14 diseases. On the task of detecting pneumonia it matched and even exceeded expert radiologists: four Stanford radiologists (4-28 years of experience) independently annotated a 420-image test set, and CheXNet significantly outperformed the average radiologist on the F1 metric. The authors further extended the model to all 14 diseases, reaching state-of-the-art results, and added Class Activation Maps (CAM) to localise and visualise the pathological regions.

Diagnosing pneumonia from an X-ray is difficult even for radiologists, because its visual signs are faint, overlap with other conditions, and mimic benign findings, producing high inter-reader variability. The task matters greatly: the chest X-ray is the primary tool for diagnosing and managing pneumonia, a disease responsible for over a million hospitalisations and roughly 50,000 deaths per year in the United States alone. Moreover, the WHO (world's health organization) estimates that two-thirds of the world's population lacks access to radiological diagnosis - a shortage of expert readers rather than of equipment - so an accurate automated reader could substantially widen access to quality diagnosis.

We chose this paper because it offers a stable, high-performing deep-learning solution to a complex medical problem, and because its architecture lends itself to domain transfer. Trained on an enormous X-ray corpus, CheXNet developed a strong visual understanding of features specific to radiographs - contrasts, tissue boundaries, and gray levels. This motivated us to test whether that understanding can transfer efficiently to a completely different body part, rather than remaining specific to the chest.

2. Proposed Extension

As our extension, we perform a domain transfer of CheXNet - originally trained on chest X-rays - to the MURA dataset of musculoskeletal radiographs, focusing on the forearm (XR_FOREARM) sub-group (for proof of concept), whose studies are labelled in binary form as abnormal (positive) or normal (negative). To isolate the contribution of the pretraining in a controlled way, we designed three experiments:

- (1) a frozen CheXNet backbone with only a new binary classifier trained .
- (2) an end-to-end fine-tuned CheXNet at a low learning rate.
- (3) the same architecture trained from scratch as a control.

We then apply Grad-CAM on the outcome of the network from experiment 2 to localise the evidence behind abnormal predictions.

Extension Reasoning

The original work stays entirely within the chest domain, whether its learned representations transfer across anatomy is not examined. We reasoned that pretraining on millions of radiographs yields useful low-level and mid-level features (edges, textures, density gradients) that are not chest-specific and should therefore benefit a different body part, while also being far more efficient than training from scratch under the compute limits of Google Colab. The extension stays tightly aligned with the original method: we keep the same DenseNet-121 backbone and replace only the final classifier with a single-output layer.

Hypothesis

Pretraining on medical X-rays, even from a different body part, produces useful inductive biases, so the CheXNet-based models (frozen and fine-tuned) will reach higher AUC-ROC and F1 and converge faster than the from-scratch model, and will yield Grad-CAM maps that are more anatomically focused on the abnormal region.

Alternatives considered

We considered a multi-class formulation predicting the specific abnormality type, and using a different backbone. We chose binary detection with the identical DenseNet-121 backbone because it isolates the transfer effect most cleanly and preserves a direct comparison to the original architecture.

3. Methodology

Architecture

All experiments use DenseNet-121 (121 convolutional layers with dense connections for efficient gradient and feature flow). Following CheXNet, we adapted the head for the binary task by replacing the 14-class classifier with a single-output Linear layer (`out_features = 1`). Under this architecture we defined three conditions:

- Experiment 1 - Frozen backbone: load CheXNet weights, freeze all layers. only the new classifier head (1,025 parameters) is trained, testing the pretrained features as-is.
- Experiment 2 - Fine-tuned backbone: load CheXNet weights, unfreeze all 6,954,881 parameters, and train end-to-end at a low learning rate to adapt to the bone domain.
- Experiment 3 - From scratch: random initialisation with no prior knowledge, serving as the control that isolates the value of CheXNet pretraining.

Data and preprocessing

We used the MURA XR_FOREARM sub-group: 1,825 training images (661 abnormal, 1164 normal) and 301 validation images (151 abnormal, 150 normal), with labels parsed from the study-folder names. A custom SquarePad transform zero-pads each image to a square before resizing, preserving the aspect ratio and preventing anatomical distortion; images are then resized to 224×224, normalised with the ImageNet mean and standard deviation (per the original paper), and augmented at training time with random rotation ($\leq 30^\circ$) and horizontal/vertical flips.

Training and tools

Each model was trained for 10 epochs with BCEWithLogitsLoss and a positive-class weight (`pos_weight` ≈ 1.761) to counter the class imbalance, optimised with Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Experiments 1 and 3 used a learning rate of 10^{-3} and Experiment 2 a lower rate of 10^{-4} to protect the pretrained representations, for each run we kept the checkpoint with the highest validation AUC-ROC. Implementation used PyTorch on a Google Colab Tesla T4 GPU, and Grad-CAM (hooked to the last DenseNet block, `denseblock4`) for interpretability.

We use AUC-ROC as the primary metric because, unlike accuracy or F1, it is threshold-independent and measures the model's ability to rank abnormal above normal across all decision thresholds rather than at a single cutoff - a more reliable indicator of discriminative quality. Accordingly, for each experiment we selected the checkpoint from the epoch with the highest validation AUC-ROC, and report that best epoch's metrics as the final results.

4. Results and Analysis

We evaluated the three runs on the validation set using AUC-ROC, Accuracy, F1, Precision, and Recall (Table 1).

Metric	Exp 1: Frozen CheXNet	Exp 2: Fine-Tuned CheXNet	Exp 3: From Scratch
AUC-ROC (best epoch)	0.613	0.865	0.693
Accuracy	0.588	0.801	0.641
F1-Score	0.637	0.774	0.614
Precision	0.571	0.896	0.667
Recall	0.722	0.682	0.570
Trainable params	1,025	6,954,881	6,954,881
Weight initialization	CheXNet (pretrained)	CheXNet (pretrained)	Random

Table 1 - Validation-set metrics, reported at the epoch with the highest validation AUC-ROC. Best value per metric in bold.

The frozen model (Exp 1) was the weakest discriminator, with the lowest AUC-ROC (0.613) and accuracy (0.588); its high recall (0.722) but low precision (0.571) reveal a poorly-calibrated head that over-predicts abnormality. Fine-tuning (Exp 2) produced a large jump to AUC-ROC 0.865, F1 0.774 and precision 0.896, showing that the chest-learned features adapt efficiently to bone once the weights are allowed to change. The from-scratch control (Exp 3) reached AUC-ROC 0.693 but trained unstably - its F1 collapsed to 0.061 at one epoch and recall swung between 0.033 and 1.000 - reflecting the difficulty of optimising a 6.95M-parameter network on $\sim 1,800$ images.

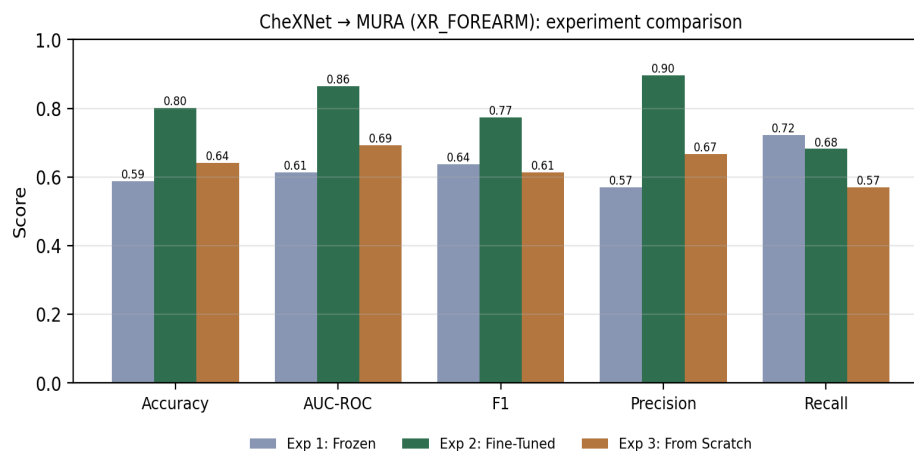


Figure 1 - Comparison of the three experiments across the five metrics.

The fine-tuned model leads the from-scratch control by about 0.172 AUC-ROC and the frozen model by about 0.252. Since all runs shared the same data, loss, and protocol, this gap demonstrates that CheXNet's pretraining supplies useful inductive biases even across anatomical domains. A notable failure case is that the frozen model (0.613) is actually beaten by training from scratch (0.693): the chest features transfer poorly when used as-is, yet serve as an excellent initialisation once fine-tuned - the pretraining steers the network away from poor local minima and yields faster, more stable convergence. The main challenge encountered was the instability of the from-scratch model on the small dataset, which we mitigated by selecting the best-AUC checkpoint.

4.1 Qualitative results - Grad-CAM

Grad-CAM localises the evidence behind an abnormal prediction, so it is meaningful only on abnormal images, we therefore applied it to true-positive cases using the best model (Exp 2). Figure

2 shows ten such cases. The fine-tuned model concentrates on the bone shaft, joints, and surgical fixation hardware - regions that are clinically relevant, since hardware marks the site of the fracture - rather than on the image borders or padding, and its confidences are high ($p \approx 0.60\text{--}0.99$). The maps are coarse (computed on a 7×7 feature map based on the last DenseNet block and up-sampled), so they indicate a region of interest rather than a pinpoint lesion. Overall, the fine-tuned model attends to anatomically relevant regions, qualitatively supporting the quantitative results.

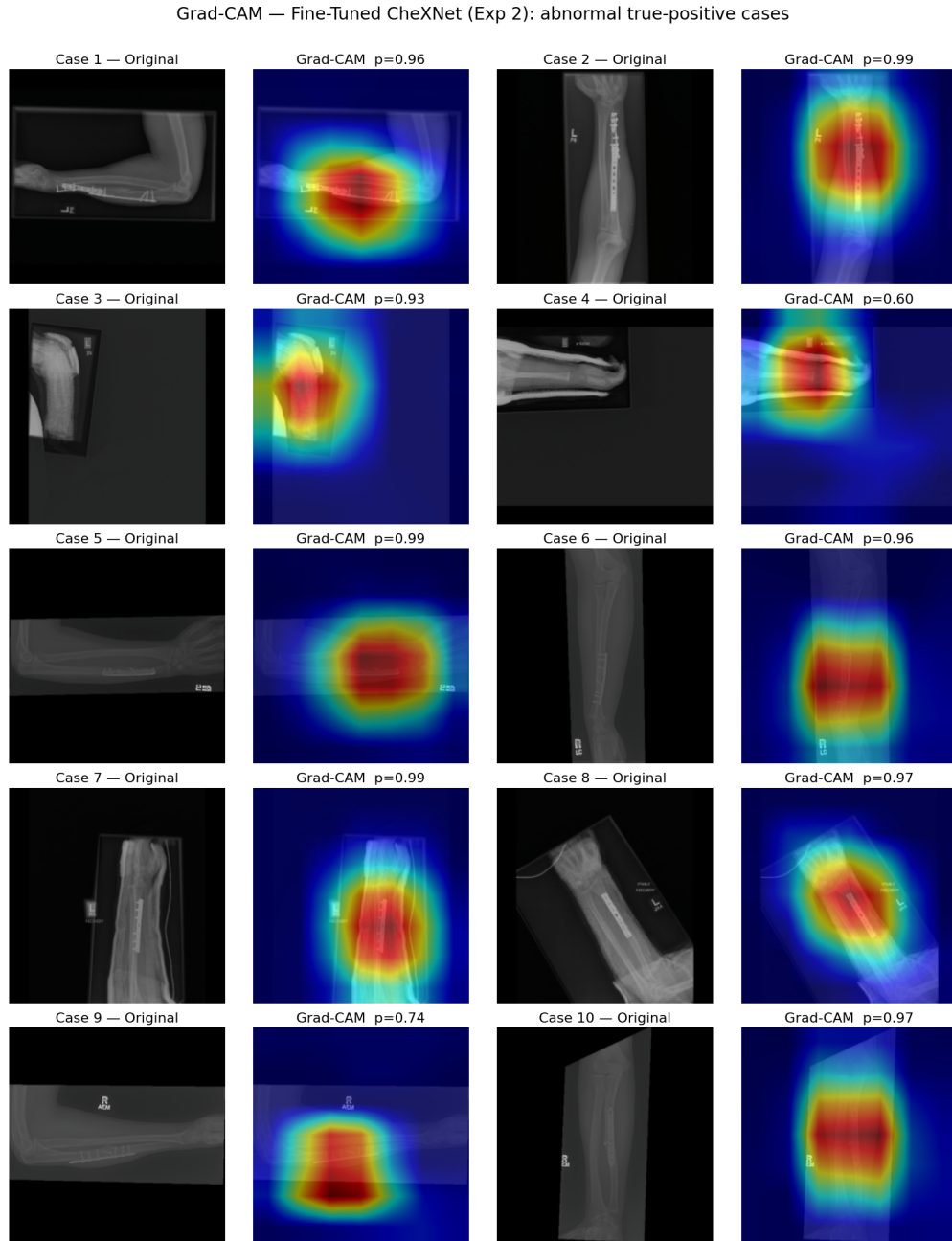


Figure 2 - Grad-CAM for the fine-tuned CheXNet (Exp 2) on true-positive abnormal forearm cases. p is the predicted probability of abnormality.

5. Conclusion

Across three controlled experiments on MURA XR_FOREARM, fine-tuned CheXNet was the clear winner (AUC-ROC 0.865, Accuracy 0.801, F1 0.774), well ahead of both the from-scratch control (AUC-ROC 0.693) and the frozen backbone (AUC-ROC 0.613). CheXNet's chest pretraining therefore provides valuable, transferable knowledge of radiographic imagery that accelerates and stabilises learning in the bone domain - but this value is realised only through fine-tuning: as a frozen feature extractor the chest representations transferred so poorly that they were beaten by random initialisation. Grad-CAM is showing that the fine-tuned model localises on anatomically relevant regions.

Limitations: The study is a proof-of-concept: it uses a single small subset (forearm, ~1,800 images), per-image rather than per-study labels, the validation set for both model selection and reporting (MURA's official test set is hidden, making the numbers mildly optimistic), and only 10 training epochs.

Future work: Natural next steps include training on the full MURA dataset and additional body parts, aggregating predictions to the study level, adding a held-out test set with early stopping, and extending from binary detection to per-abnormality classification. Overall, our results indicate that transfer learning from large medical-imaging corpora is worthwhile even across anatomical domains - provided the model is fine-tuned rather than used as a frozen feature extractor.